Experimental and method

LOGICLAB KIDS · EUCYS 2026

Native Platforms
01

Native **SwiftUI (MVVM)** on iOS (live on Apple TestFlight) and **Kotlin Compose** on Android. Operates offline-first; all lessons execute with zero network connectivity.

Stage Grammar
02

6 modules and 36 lessons bundled as JSON. 4 interactive task types: Quiz, Spot-It, Sort-It, Build-It. Mistakes are never punished (explanations shown after 2 misses).

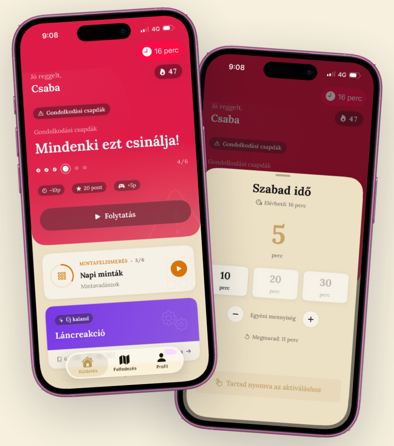
Premack Contingency
03

Applies the Premack (1959) differential probability law. Solving a lesson credits the ledger with **+20 pts & +5 min**. Child spends balance within parent-allowed hours.

Constrained Gemini
04

System prompt restricts Gemini to Socratic dialogue: probes reasoning and gives hints, never direct answers. Strict low token cap prevents lecturing; unlocks only post-task.

Interactive Child Experience · iOS & Android



01 · DAILY MYSTERY

02 · EARN FREE TIME

The Behavioral Lock

Screen entertainment is the highest-probability behavior; structured reasoning practice is among the lowest. The lock and key are in the OS: a child cannot negotiate around it.

System Architecture

OS Sandbox vs Cloud

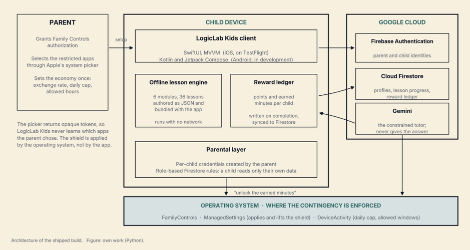


FIG 3. Hardware sandbox vs cloud: Screen-time contingency is enforced on-device via Apple FamilyControls and ManagedSettings. Cloud Firestore manages sync and parental rules.

Stage Grammar & Loop

5-Stage Interaction Cycle

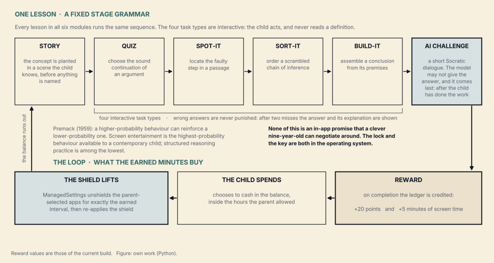


FIG 4. Fixed 5-stage lesson grammar (Story → Quiz/Spot/Sort/Build → Socratic AI → Reward Credit → Shield Lift). Screen entertainment is unlocked strictly for the earned interval.

Opaque Token Shielding

The system picker returns **opaque tokens**, so LogicLab Kids never accesses which apps the parent selected. Shields are applied directly by iOS ManagedSettings, not in-app.

4 Core Task Types

Quiz (sound continuation), **Spot-It** (faulty step), **Sort-It** (inference chain), and **Build-It** (conclusion assembly). The child acts dynamically rather than memorizing definitions.

36	+5 min	100%	0 Net
Interactive Lessons	**Earned Screen Time**	**OS-Level Shielding**	**Offline Runtime**
6 modules across 4 tracks	+20 pts per solved lesson	Apple FamilyControls & Opaque Tokens	Zero network required for lessons

REFERENCES

[1] Apple Inc. FamilyControls, ManagedSettings & DeviceActivity Frameworks.

[2] Premack, D. (1959). Toward empirical behavior laws: I. Positive reinforcement.

[3] Kosmyna et al. (2025). Your Brain on ChatGPT. *MIT Media Lab*.

[4] LogicLab Kids System Architecture & Build Specification (2026).